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| Exp No: 9 | Implementation of SUBNETTING in CISCO Packet Tracer Simulator |
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## AIM:

To design multiple subnets with a suitable number of hosts, assign static IP addresses, connect the subnets through a router in Cisco Packet Tracer, and observe packet transmission with and without the router.

## ALGORITHM / PROCEDURE:

1. Take the network address 192.168.1.0/24 and decide the number of subnets needed (3 departments, so 4 subnets are created by borrowing 2 host bits).
2. The new mask is 255.255.255.192 (/26). Each subnet has 64 addresses, so 62 usable hosts.
3. Open Packet Tracer and place 1 router (2911), 3 switches (2960) and 6 PCs. Connect them with copper straight-through cables as in Fig 1.
4. Assign a static IP address, subnet mask and default gateway to every PC under *Desktop > IP Configuration*.
5. Configure the router interfaces: `interface gig0/0`, `ip address 192.168.1.1 255.255.255.192`, `no shutdown`
6. Repeat step 5 for Gig0/1 and Gig0/2 with the gateway address of subnets B and C.
7. From the PC command prompt use `ping` to test connectivity inside a subnet and across subnets.
8. Remove the router (delete the links to the router) and ping across the subnets again to see the difference.

**Fig 1: Network topology with three subnets joined by a router**

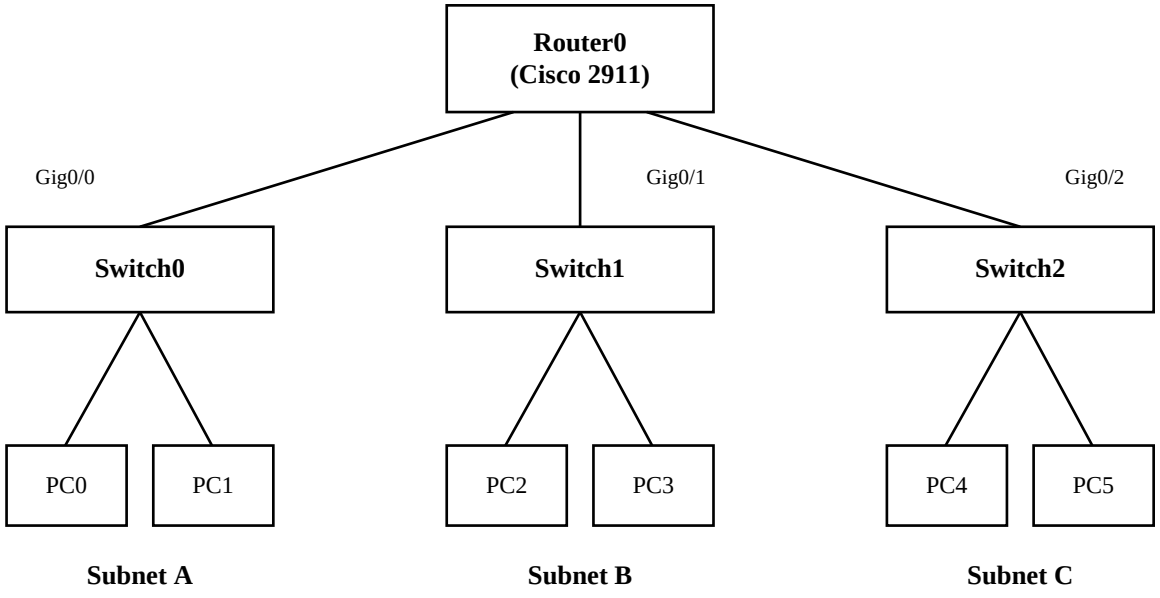
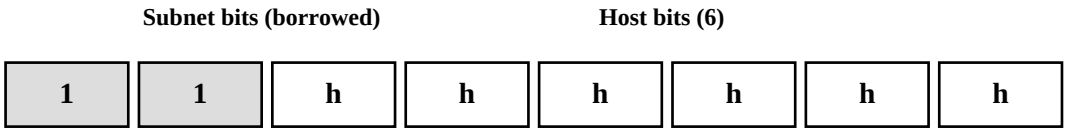


**Fig 2: Borrowing 2 bits of the host part for subnetting**




192.168.1.0 /24 -> /26 (2 bits borrowed = 4 subnets, 2^6 - 2 = 62 hosts each)

## Subnet design:

| Subnet | Network ID | Mask | Usable range | Broadcast | Gateway |
|---|---|---|---|---|---|
| A (Sales) | 192.168.1.0/26 | 255.255.255.192 | .1 - .62 | 192.168.1.63 | 192.168.1.1 |
| B (HR) | 192.168.1.64/26 | 255.255.255.192 | .65 - .126 | 192.168.1.127 | 192.168.1.65 |
| C (IT) | 192.168.1.128/26 | 255.255.255.192 | .129 - .190 | 192.168.1.191 | 192.168.1.129 |
| Unused | 192.168.1.192/26 | 255.255.255.192 | .193 - .254 | 192.168.1.255 | - |

## IP address table:

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| Router0 | Gig0/0 | 192.168.1.1 | 255.255.255.192 | - |
| Router0 | Gig0/1 | 192.168.1.65 | 255.255.255.192 | - |
| Router0 | Gig0/2 | 192.168.1.129 | 255.255.255.192 | - |
| PC0 | Fa0 | 192.168.1.2 | 255.255.255.192 | 192.168.1.1 |
| PC1 | Fa0 | 192.168.1.3 | 255.255.255.192 | 192.168.1.1 |
| PC2 | Fa0 | 192.168.1.66 | 255.255.255.192 | 192.168.1.65 |
| PC3 | Fa0 | 192.168.1.67 | 255.255.255.192 | 192.168.1.65 |
| PC4 | Fa0 | 192.168.1.130 | 255.255.255.192 | 192.168.1.129 |
| PC5 | Fa0 | 192.168.1.131 | 255.255.255.192 | 192.168.1.129 |

## Router configuration:

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.1.129 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#end
Router#write memory
```

## OUTPUT:

Router0 CLI - show ip interface brief

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.1.65    YES manual up                    up
GigabitEthernet0/2     192.168.1.129   YES manual up                    up
```

PC0 Command Prompt - same subnet

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

### (a) When the subnets are connected via a router

PC0 Command Prompt - ping PC2 (Subnet B) through the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time=3ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router

```
PC>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=2ms TTL=127
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router (contd.)

```
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

**(b) When the subnets are not connected (router removed)**

PC0 Command Prompt - ping PC2 (Subnet B) without the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

PC0 Command Prompt - ping PC1 (same subnet) without the router

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

## OBSERVATION:

| Case | Source -> Destination | Result |
|---|---|---|
| Router connected | PC0 -> PC1 (same subnet) | Success, TTL = 128 |
| Router connected | PC0 -> PC2 / PC4 (other subnet) | Success, TTL = 127 (one router hop). First packet lost while ARP is resolved. |
| Router removed | PC0 -> PC1 (same subnet) | Success, switch forwards the frames |
| Router removed | PC0 -> PC2 / PC4 (other subnet) | Fails, 100% loss. No device can route between subnets. |

## RESULT:

Thus subnetting of the 192.168.1.0/24 network into /26 subnets was implemented in Cisco Packet Tracer. Hosts in different subnets could communicate only when a router was connected between them, while hosts inside the same subnet could communicate even without the router.